

Writing Automated Tests for Exception Throwing Code



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Module Overview

Testing exceptions with NUnit

Testing exceptions with xUnit.net

Testing exceptions with MSTest V2

Key course takeaways and resources



Module Summary

Testing exceptions with NUnit

- Assert.That(... , Throws.TypeOf<...>());
- Assert.That(... , Throws.InstanceOf<...>());

Testing exceptions with xUnit.net

- Assert.Throws<...>(...)
- Assert.ThrowsAny<...>(...)

Testing exceptions with MSTest V2

- Assert.ThrowsException



Key Takeaways

Use exceptions not error return codes

Thrown exceptions bubble up the call stack until they get caught by a catch block or crash to the operating system

All exceptions inherit from `System.Exception`

Exceptions can wrap other exceptions

Use try & catch blocks to handle exceptions

Use a finally block to execute clean-up code



Key Takeaways

Do not add a catch block that does nothing or just rethrows – catching should add value

Do not use exceptions for normal program flow logic

Be careful to preserve the stack trace when rethrowing

Only create custom exceptions when existing exception types are not suitable

Avoid sensitive information in exceptions as exceptions could be logged



Resources and Further Learning

C# programming guide to exceptions and exception handling

<https://bit.ly/excsguide>

Framework design guidelines for exceptions

<https://bit.ly/exframe>



Resources and Further Learning

Testing .NET Code with xUnit.net: Getting Started

<https://bit.ly/xunitcourse>

Automated Testing with MSTest V2

<http://bit.ly/psmstest>

Introduction to .NET Testing with NUnit 3

<https://bit.ly/nunitcourse>

